TP5

Mohamed Ali Knis - TD3/TP6

```
import kagglehub
from kagglehub import KaggleDatasetAdapter
import matplotlib.pyplot as plt

df = kagglehub.dataset_load(
    KaggleDatasetAdapter.PANDAS,
    "azeembootwala/titanic",
    "test_data.csv",
)


# Partie 1
print(df.head())
print(df.shape)
print(df.isnull().sum())

print(df["Sex"])
df["Age"] = df["Age"].fillna(df["Age"].mean())
print(df["Sex"])

print(df["Sex"])
df["Sex"] = df["Sex"].map({0: "male", 1: "female"})
print(df["Sex"])

# Partie 2

# histogramm
df["Age"].plot(kind="hist")
plt.show()

# barplot
df["Sex"].value_counts().plot(kind="bar")
plt.show()

# Boxplot
df.boxplot(column="Fare")
plt.show()


# encodage de Pclass(necessaire car my dataset structure et different donc il faut fair un Transformation)
def get_class(row):
    if row["Pclass_1"] == 1:
        return "1"
NORMAL  TP5.py  python                                              utf-8  python  53%  38:1
"TP5.py" 71L, 1561B written
```

```

# Boxplot
df.boxplot(column="Fare")
plt.show()


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def get_class(row):
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    if row["Pclass_2"] == 1:
        return "2"
    if row["Pclass_3"] == 1:
        return "3"
    return "Inconnu"

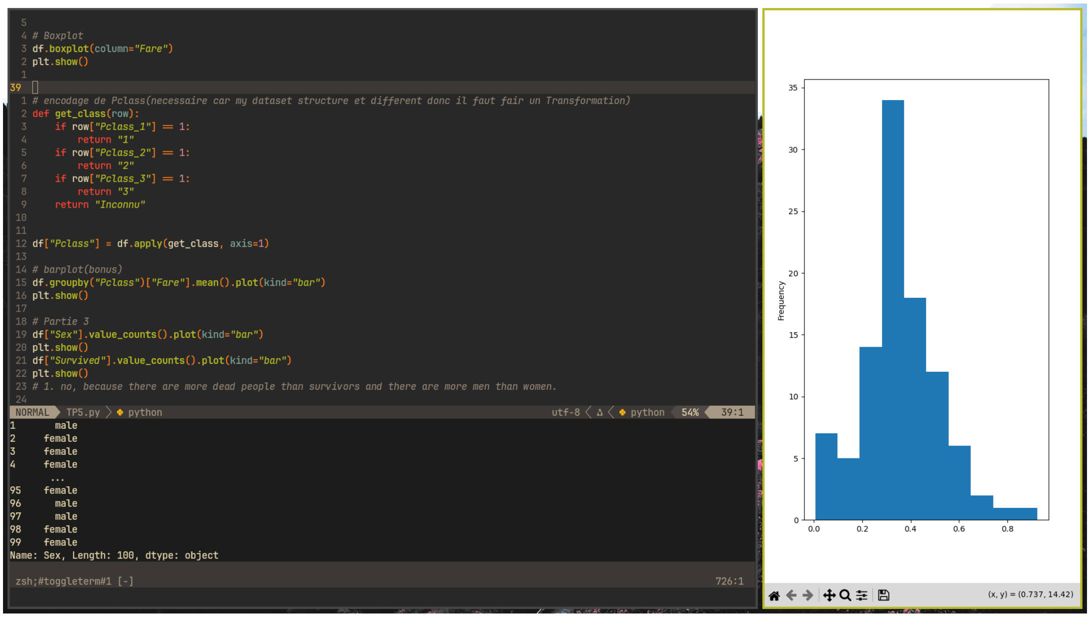

df["Pclass"] = df.apply(get_class, axis=1)

# barplot(bonus)
df.groupby("Pclass")["Fare"].mean().plot(kind="bar")
plt.show()

# Partie 3
df["Sex"].value_counts().plot(kind="bar")
plt.show()
df["Survived"].value_counts().plot(kind="bar")
plt.show()
# 1. no, because there are more dead people than survivors and there are more men than women.

df.boxplot(column="Fare")
plt.show()
# 2. oui, because the boxplot show the extreme values

df.groupby("Pclass")["Fare"].mean().plot(kind="bar")
plt.show()
print(df.groupby("Pclass")["Fare"].mean())
# 3. class 1 paie le plus avec un cout moyenne plus que le somme de le deux autre classes combine.
NORMAL  TP5.py  python                                              utf-8  python  54%  39:1
"TP5.py" 71L, 1561B written
```



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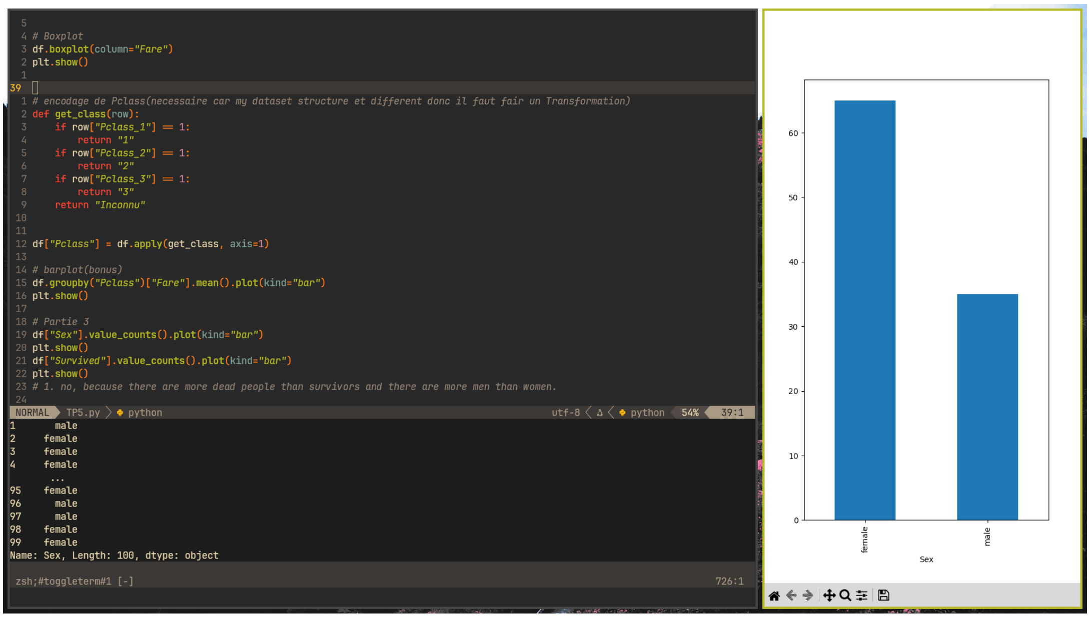
NORMAL  TP5.py  python                                utf-8  python  54%  39:1
1       male
2     female
3     female
4     female
       ...
95    female
96      male
97      male
98    female
99    female
Name: Sex, Length: 100, dtype: object

zsh;#toggleterm#1 [-]                                                    726:1
```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zsh;#toggleterm#1 [-]                                                    726:1
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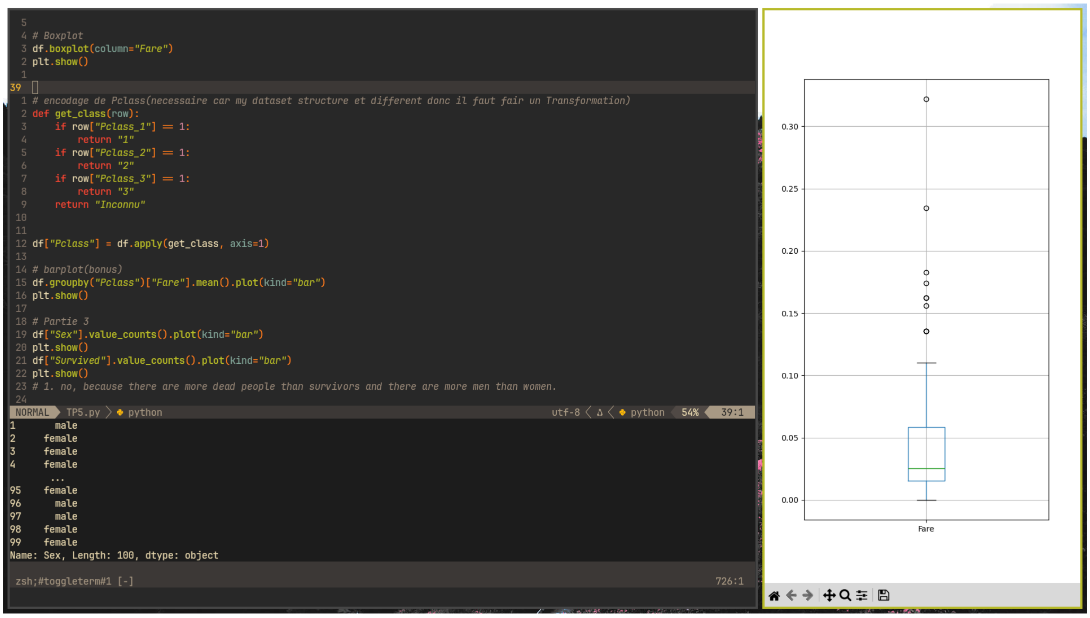

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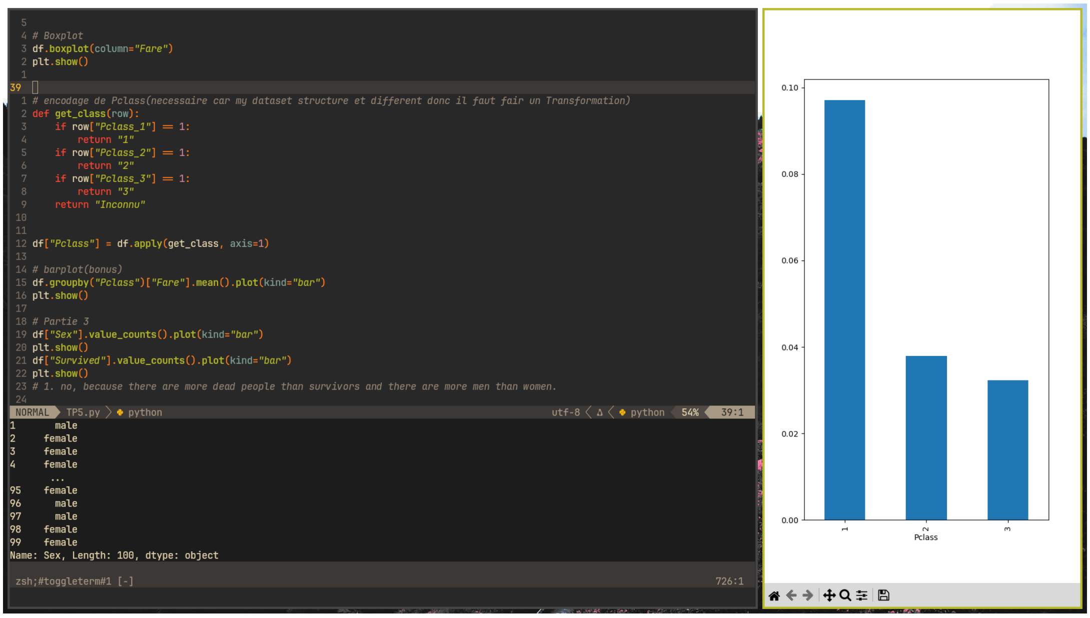
1 male
2 female
3 female
4 female
...
95 female
96 male
97 male
98 female
99 female
Name: Sex, Length: 100, dtype: object
zsh;#toggleterm#1 [-] 726:1
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Fare



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2 female
3 female
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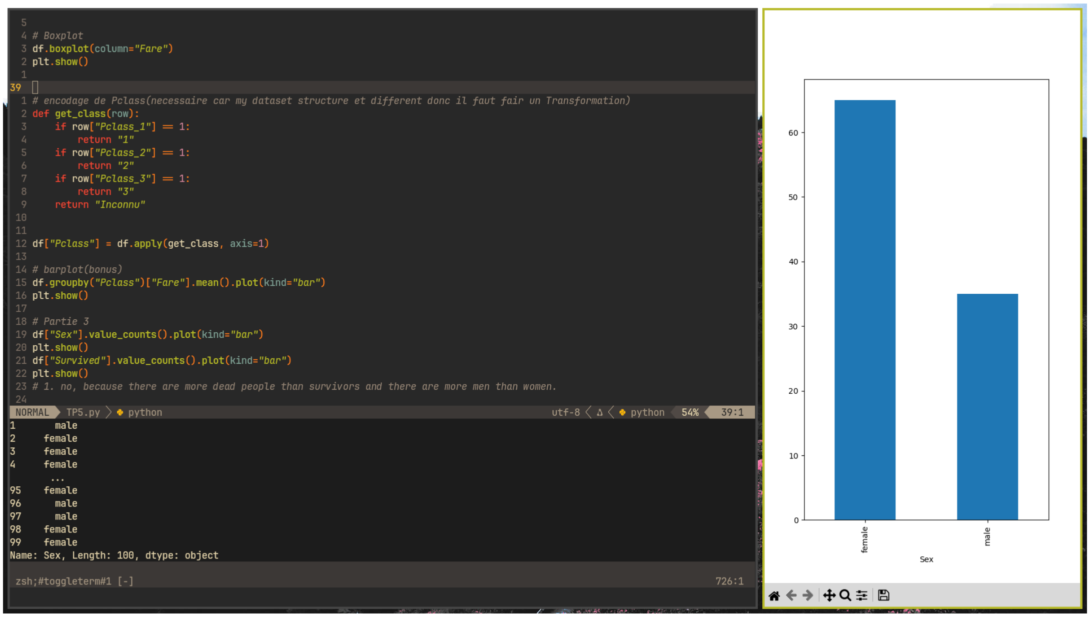
95 female
96 male
97 male
98 female
99 female
Name: Sex, Length: 100, dtype: object
zsh;#toggleterm#1 [-] 726:1
0.10
0.08
0.06
0.04
0.02
0.00
1
2
3
Pclass



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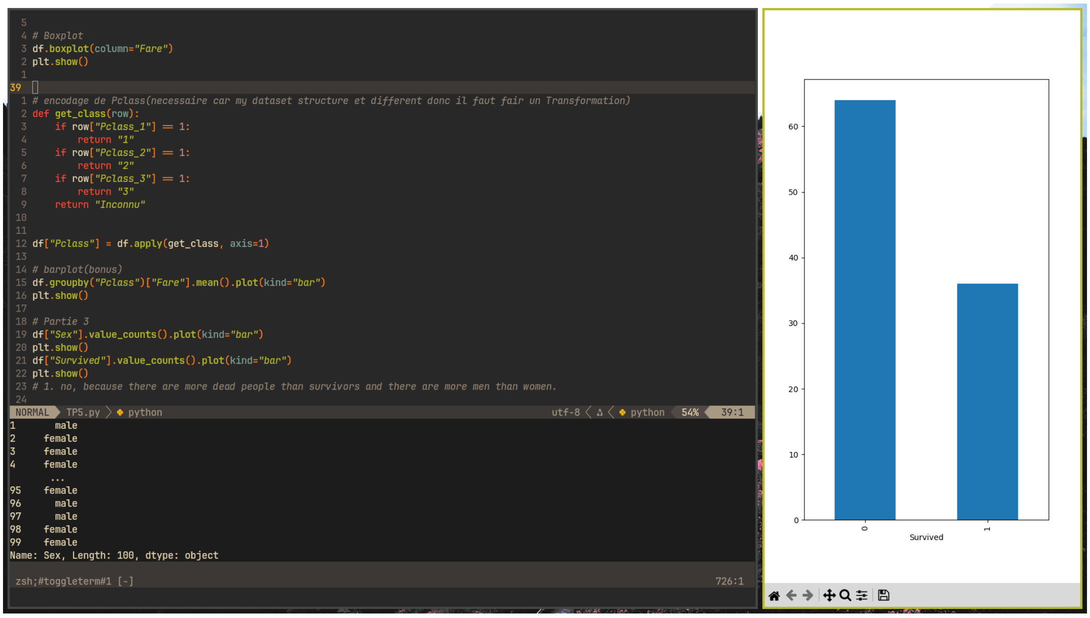
zsh;#toggleterm#1 [-] 726:1
60
50
40
30
20
10
0
female
male
Sex



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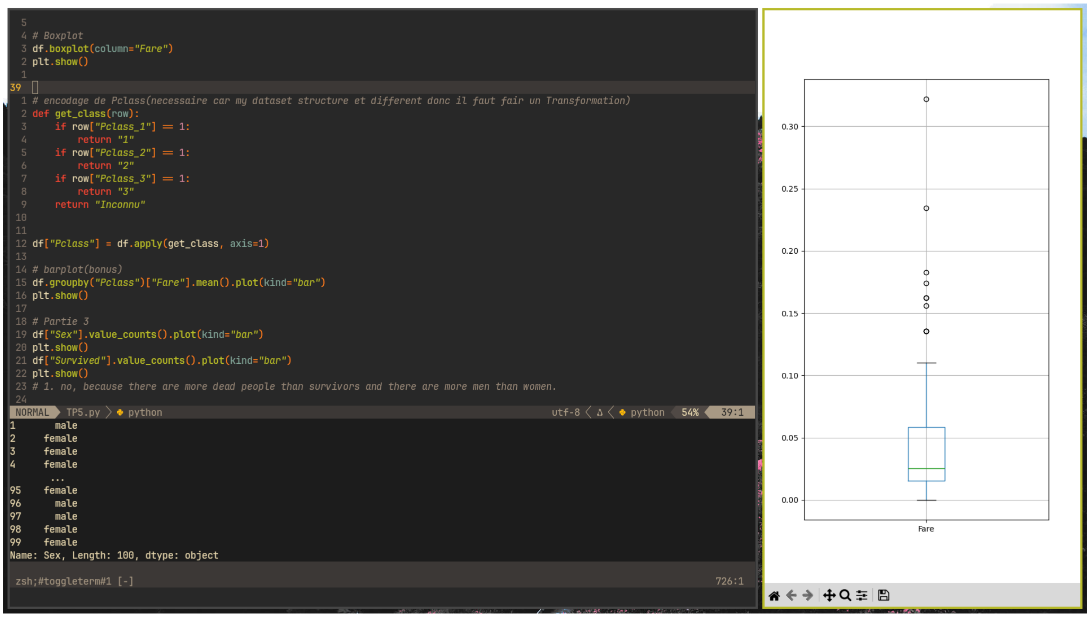
zsh;#toggleterm#1 [-] 726:1
60
50
40
30
20
10
0
0
1
Survived



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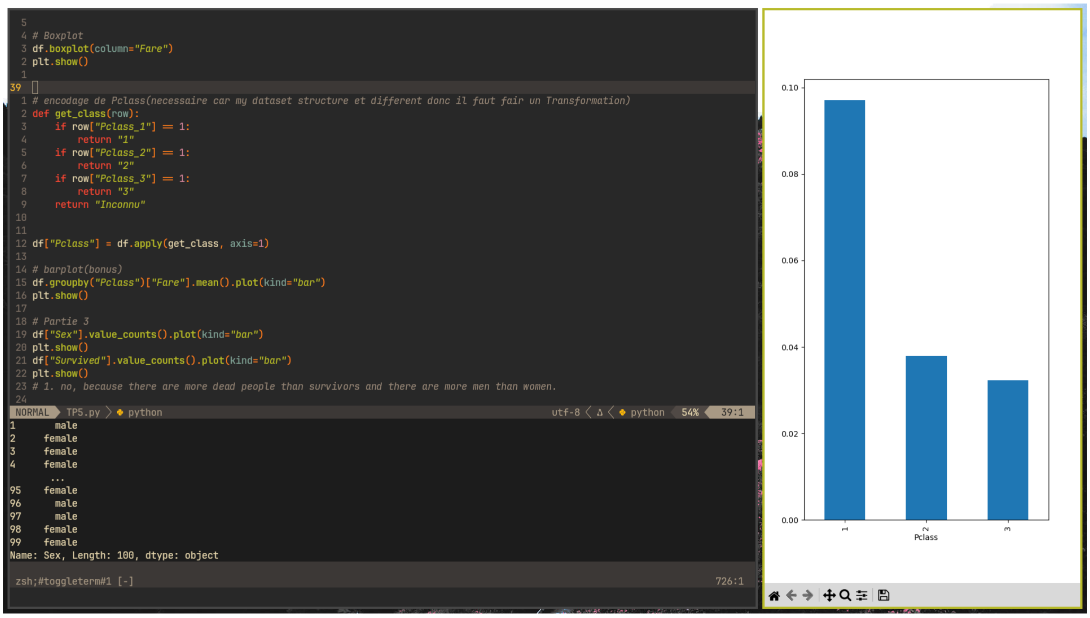

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0.10
0.08
0.06
0.04
0.02
0.00
1
2
3
Pclass

```python

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```

```
NORMAL  TP5.py  python                                    utf-8  python  54%  39:1
97      male
98    female
99    female
Name: Sex, Length: 100, dtype: object
Pclass
1    0.097096
2    0.037951
3    0.032300
Name: Fare, dtype: float64

Dev/BI/TP5 via 🐍 v3.13.12 via ❄  impure (nix-shell-env) took 16s
10% ❯
 zsh;#toggleterm#1 [-]                                                    733:1
```